CSE2200: DESIGN PROJECT - II

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Project Title: AgroInvest: A Crowdfunding Platform for Farmers and Investors

Problem Identification

Agriculture is one of the most important sectors in Bangladesh. However, many farmers own agricultural land but do not have enough financial resources to cultivate their crops. They often struggle to purchase seeds, fertilizers, pesticides, irrigation equipment, and labor due to a lack of capital.

At the same time, many individuals are interested in investing small amounts of money but do not have a transparent and organized platform for agricultural investment.

To address this problem, AgroInvest is proposed as a web-based crowdfunding platform where verified farmers can create farming projects and request funding from multiple investors. Investors can browse approved projects, invest through SSLCommerz Sandbox, and track project progress through funding updates, milestone tracking, and farmer trust scores. The platform also calculates the expected profit or loss distribution for academic demonstration purposes.

Project Objectives

The main objectives of the system are:

- To provide a digital platform where farmers can create agricultural funding projects.
- To enable investors to invest small amounts in farming projects.
- To display transparent funding progress for every project.
- To provide project milestone updates throughout the farming cycle.
- To maintain a farmer trust score based on completed projects and investor ratings.
- To integrate SSLCommerz Sandbox for investment simulation.
- To calculate and display profit/loss distribution.
- To improve transparency and encourage digital agricultural investment.

Scope of the Project

In Scope

- Farmer Registration and Login
- Investor Registration and Login
- Admin Login

- Farmer Project Creation
- Admin Project Approval
- Project Browsing
- Funding Progress Tracking
- SSLCommerz Sandbox Payment Integration
- Investment History
- Project Milestone Tracking
- Farmer Trust Score
- Profit/Loss Calculation (Simulation)
- Admin Dashboard
- REST API Integration
- Testing & QA
- Deployment

Out of Scope

- Real money payout to investors
- Direct bank transfer to farmers
- Legal contract management
- Crop insurance services
- IoT-based farm monitoring
- Government subsidy management
- Real-time satellite or drone monitoring

Feasibility Study

Technical Feasibility

- The proposed system can be developed using HTML, CSS, Bootstrap, JavaScript, PHP, MySQL, REST API, and SSLCommerz Sandbox**.** These technologies are widely available, easy to learn, and suitable for software engineering projects. The development team has the required knowledge to implement the system.

Economic Feasibility

- The development cost is low because open-source technologies will be used. SSLCommerz Sandbox provides a free testing environment without real financial transactions. The project can also be deployed using low-cost hosting services.

Operational Feasibility

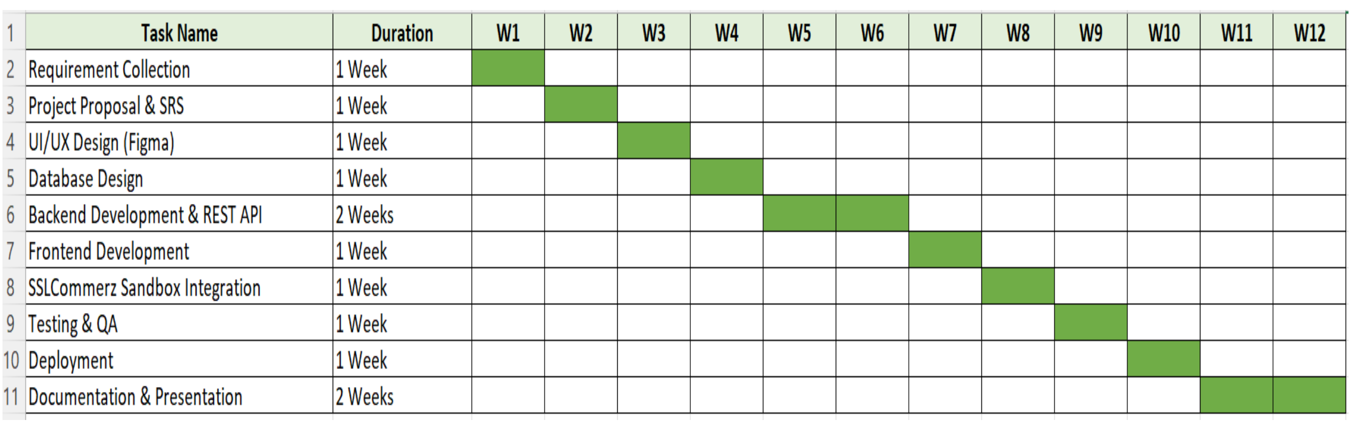
- The system is designed for verified farmers, investors, and administrators. Farmers can create projects and update milestones, investors can browse and invest in projects, and administrators can approve projects and monitor activities. Farmers who are not familiar with digital technology may receive assistance from family members or local digital service centers.

Schedule Feasibility

- The project is expected to be completed within 10–12 weeks following proper project planning, system design, development, testing, and deployment.

Project Schedule

The following project schedule presents the planned timeline for developing the **AgroInvest** system over 12 weeks. It includes all major phases of the project, from requirement collection and system design to development, testing, deployment, and final documentation.



1	Task Name	Duration	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
2	Requirement Collection	1 Week												
3	Project Proposal & SRS	1 Week												
4	UI/UX Design (Figma)	1 Week												
5	Database Design	1 Week												
6	Backend Development & REST API	2 Weeks												
7	Frontend Development	1 Week												
8	SSLCommerz Sandbox Integration	1 Week												
9	Testing & QA	1 Week												
10	Deployment	1 Week												
11	Documentation & Presentation	2 Weeks												

Conclusion

AgroInvest is an innovative and practical web-based crowdfunding platform designed to support the agricultural sector. By connecting verified farmers with investors through a secure and transparent digital platform, the system encourages agricultural financing while demonstrating modern software engineering practices. The project is feasible to implement within the academic timeline and fulfills the required software engineering components, including Project Management, Git, UI/UX, REST API, Payment Gateway, Testing, and Deployment.